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DESIGN OF A LOGICAL WASTE COLLECTION AND COMMUNITY REPORTING SYSTEM FOR FREETOWN (ABSTRACT)

Another challenge facing Freetown is the management of waste. Every day, waste is produced by the various households, businesses, markets, schools, offices, restaurants and public institutions within the city. To effectively manage waste, a system must be established to report, collect, transport and manage the waste. If the system is inefficient, the waste will remain in the community for extended periods and create problems for both the residents and the environment.

In this project, however, the problem of waste is not just the existence of waste. Rather, the problem is how to report, record and assign waste-related problems to the appropriate service area within the city. The severity of the waste problem can vary greatly depending on the location of the waste. For instance, if there is only a small amount of waste in an ordinary location, it will likely require the normal response from the waste management department. However, if there is waste accumulating near a drainage channel, market or area with a high population density, it will require a more urgent response from the waste management department. If all reports of waste are treated in the same way, then the waste management resources may not be properly allocated to the areas that require the most attention.

In addition to these challenges, another issue is that the information regarding the waste problem may be incomplete or inconsistent. For instance, a report may state that there is too much rubbish in an area but does not provide any further details as to where the problem exists, what type of waste is involved, for how long the waste has been present in the area, how many people are affected by the problem or if there is any environmental risk associated with the waste problem.

The purpose of this project is to create a logical method for transforming a community waste complaint into structured information.

Background of the Problem

Freetown's a bustling city—tons of people living, working, studying, and running businesses here every day. With all that activity, you get a whole mix of waste piling up everywhere. Now, waste doesn't just show up once and disappear after pickup. The stuff keeps coming, so the waste-management system has to keep running nonstop. The only way to really deal with this is for everyone—community members, those producing the waste, city authorities, and the folks collecting it—to work together consistently.

Freetown City Council set up some rules and frameworks for handling waste and keeping things clean. These aren't just guidelines—they actually lay out how waste-collection services operate across the city. Because there are specific areas and service providers, you can pin down waste issues to certain spots pretty easily.

And that's where programming logic steps in. Instead of seeing waste management as one huge, overwhelming task, it helps to break things down into smaller chunks. So, say someone in the community notices a problem—they report it. The system checks if their report has all the info it needs. If it passes, then the system sorts out what kind of problem it is and decides how urgent it is—low, medium, high, or urgent.

The project's roots go even deeper, tying back to Freetown's environmental situation. Waste and drainage in the city are intertwined. If garbage ends up in or near the drainage channels, it can cause bigger environmental headaches, especially when it rains hard. That means when the system logs a report about waste, it shouldn't just mark it down—it should look at what makes that report more pressing than others.

In the end, this project shows how applying programming logic can make it easier to manage and organize waste information—helping the city handle the mess in a smarter, more efficient way.

FREETOWN COMMUNITY WASTE REPORTING AND COLLECTING SYSTEM

The proposed solution is a Freetown Community Waste Reporting and Collection Prioritization System

The system would allow a waste problem to be entered as a structured report. The report would then pass through several logical stages.

First, the system receives information about the waste problem. It then checks whether the required information has been provided. If information is missing, the report is returned for correction.

If the information is complete, the system analyses the report. It identifies the type and severity of the problem and checks important conditions such as whether the waste is near a drainage channel or whether there is a potential health or flooding concern.

The system then calculates a priority level. For example, reports involving significant environmental or community risks could receive higher priority than ordinary reports.

After the priority has been determined, the system identifies the relevant waste-management area and assigns the report to the appropriate service queue.

The report can then move through different statuses:

Pending → Assigned → In Progress → Resolved

This creates a simple logical tracking process.

The proposed system can therefore be represented as:

Report → Validate → Classify → Calculate Priority → Determine Area → Assign → Monitor → Resolve

COMPUTATIONAL PROBLEM SOLVING

ALGORITHMIC DESIGN OF THE LOGICAL WASTE COLLECTING AND COMMUNITY REPORTING SYSTEM

- Start
- Receive a waste report.
- Enter the report information.
- Validate the information.
- If required information is missing, display an error.
- If the information is valid, continue processing.
- Identify the waste type and location.
- Calculate a priority score.
- Determine the priority category.
- Identify the responsible waste-management area.
- Assign the report.
- Add the report to the collection queue.
- Monitor the status.
- If collection is completed, mark the report as resolved.
- If collection has not been completed, keep the report pending or in progress.
- Generate the relevant output.
- End.

A community report would contain information such as:

- Report ID
- Location
- Waste type
- Number of days waiting
- Whether the location is near a drain
- Whether there is flood risk
- Whether there is a health risk

The system first checks whether the required information is valid. It then calculates a priority score using the project's designed scoring model.

The system could then classify the result as:

Score	Priority
12 or above	Urgent
9-11	High
5–8	Medium
Below 5	Low

PSEUDOCODE DESIGN OF THE WASTE MANAGEMNET AND REPORT SYSTEM

The pseudocode demonstrates:

- Input of waste-report information;
- Validation of information;
- Variable use;
- Boolean logic;
- Arithmetic calculations;
- Decision/selection structures;
- Repetition;
- Assignment of values;
- Output generation;
- Status monitoring.

The overall processing sequence is:

```
BEGIN
    INPUT Location
    INPUT WasteType
    INPUT DaysWaiting
    INPUT NearDrain
    INPUT FloodRisk
    INPUT HealthRisk
     IF  Location is not empty AND
       WasteType is not empty AND
       DaysWaiting >= 0 THEN
        PriorityScore ← 0
        IF FloodRisk = TRUE THEN
            PriorityScore ← PriorityScore + 5
        END IF
        IF HealthRisk = TRUE THEN
            PriorityScore ← PriorityScore + 5
        END IF
        IF NearDrain = TRUE THEN
            PriorityScore ← PriorityScore + 4
        IF PriorityScore >= 14 THEN
            PriorityLevel ← "URGENT"
        ELSE IF PriorityScore >= 10 THEN
            PriorityLevel ← "HIGH"
        ELSE IF PriorityScore >= 5 THEN
            PriorityLevel ← "MEDIUM"
        ELSE
            PriorityLevel ← "LOW"
        END IF
        Determine WasteBlock
        Assign Provider
        Status ← "Pending"
        INPUT CollectionCompleted
        IF CollectionCompleted = TRUE THEN
            Status ← "Resolved"
        ELSE
            Status ← "In Progress"
```

```
        END IF
        OUTPUT ReportID
        OUTPUT PriorityScore
        OUTPUT PriorityLevel
        OUTPUT WasteBlock
        OUTPUT Provider
        OUTPUT Status
    ELSE
        OUTPUT "INVALID REPORT INFORMATION"
END
```

A FLOWCHART DEPICTION OF THE FRAMEWORK OF THE SOFTWARE

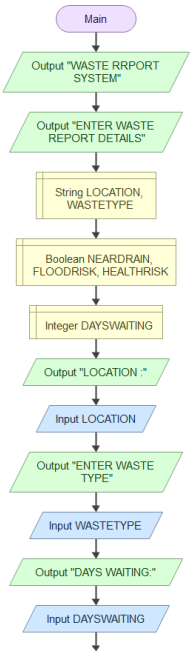
Purpose of the Input Section

The input section is the first major stage of the Waste Collection and Community Reporting System. Its purpose is to collect the information required to understand and process a reported waste problem.

The system receives:
LOCATION
WASTETYPE
DAYSWAITING
NEARDRAIN
FLOODRISK
HEALTHRISK

These represent the inputs required to process a report.

In the proposed system, a report should not simply state that "there is waste." The system needs several pieces of information so that it can determine the nature and urgency of the problem.



PROCESSING SECTION OF THE FLOWCHART

The processing section is where the system takes the input information and performs operations on it.
The inputs are not useful by themselves. The processing stage transforms them into meaningful information.

Initialize Priority Score

After validation, the first processing operation should be:

```
PRIORITYSCORE = 0
```

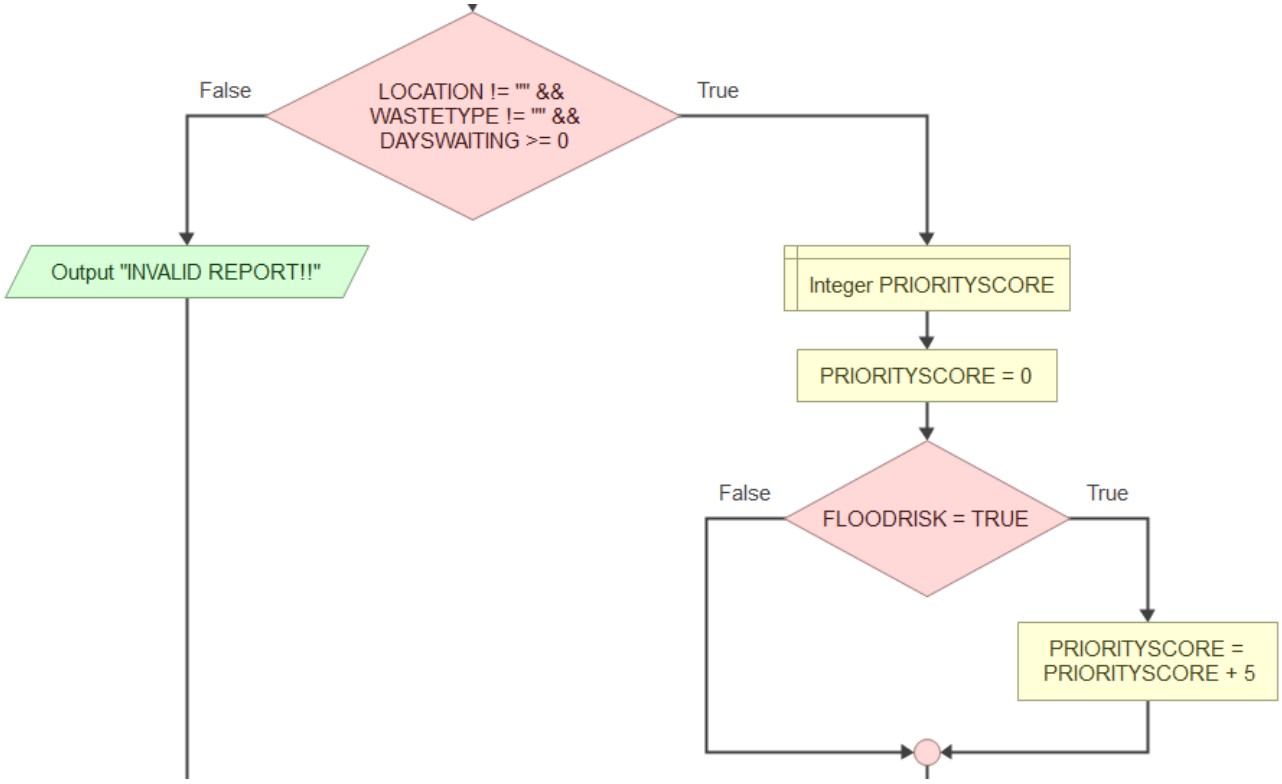
This is important because every new report needs to begin with a fresh score.
Without resetting the variable, a previous report's score could theoretically affect the next report.
For example:

```
Report 1 → Score = 17
```

When Report 2 starts, the system must reset:

```
PRIORITYSCORE = 0
```

before calculating Report 2.



Priority Model

proposed model is:

Condition		Points
Flood risk		+5
Health risk		+5
Near drain		+4

Example of the Priority Calculation

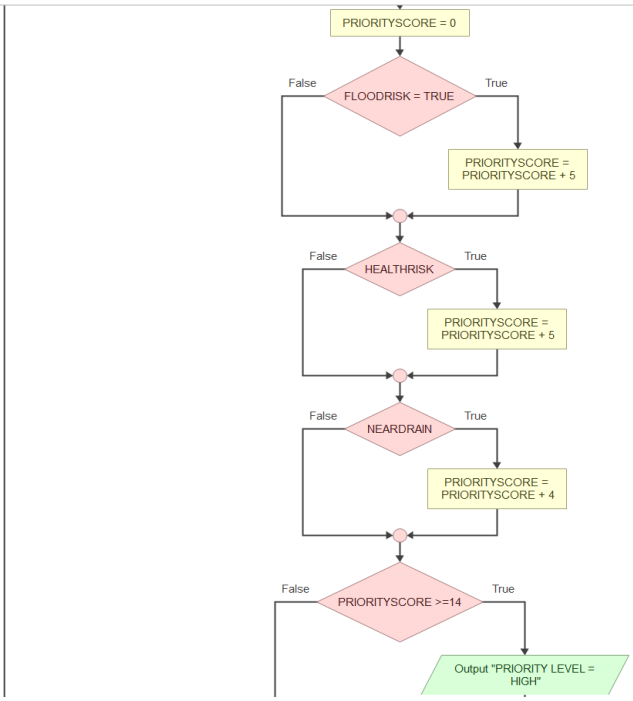
Suppose a report has:

Flood Risk = TRUE
Health Risk = TRUE
Near Drain = TRUE
Quantity Level = High
People Affected = 4
Days Waiting = 3

The calculation becomes:

Flood Risk = 5
Health Risk = 5
Near Drain = 4

Total = 14



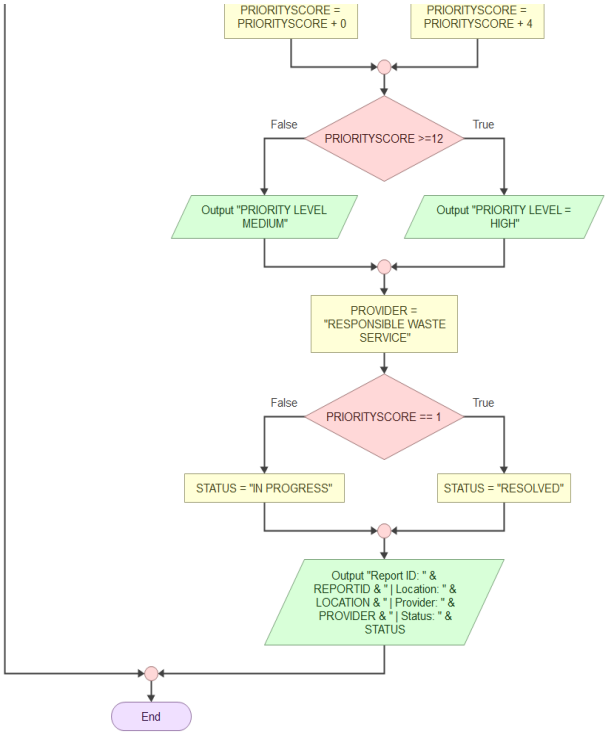
Output and Reporting

The output stage presents the results generated by the Waste Collection and Community Reporting System for Freetown. After a waste report has been validated, prioritised and assigned to the responsible waste service provider, the system displays the report details to help users and management monitor its progress.

The system produces the following information:

1. Report ID: A unique identification number automatically generated for each new report, such as R1, R2 and R3.
2. Location: The community or area where the waste problem was reported.
3. Priority Score: The numerical score calculated from factors such as flood risk, health risk, proximity to drains, quantity of waste, number of people affected and waiting time.
4. Priority Level: The urgency category assigned to the report — URGENT, HIGH, MEDIUM or LOW.
5. Assigned Provider: The waste service provider responsible for handling the report.
6. Collection Status: Indicates whether the report is assigned, in progress or resolved.

THE OUTPUT SECTION



Sustainable Development Goal (SDG) Alignment

The proposed Waste Collection and Community Reporting System for Freetown is closely aligned with the United Nations Sustainable Development Goals (SDGs) because the project addresses an environmental and community-management problem through a structured digital solution.

The primary SDG for this project is SDG 11: Sustainable Cities and Communities. SDG 11 focuses on making cities and human settlements inclusive, safe, resilient and sustainable. More specifically, Target 11.6 calls for reducing the environmental impact of cities, including through better attention to municipal and other waste management.

This makes SDG 11 particularly relevant to the proposed system because the system is designed around the reporting, prioritisation, assignment and monitoring of waste-related problems.

How the project contributes to SDG 11

The system can support SDG 11 in several ways:

1. Improved waste reporting

Residents and other community members can provide information about waste problems instead of problems remaining unrecorded or being communicated inconsistently.

For example, a community member could report:

- Location of accumulated waste
- Type of waste
- Approximate quantity
- Number of people affected
- Number of days the problem has existed
- Whether the waste is near a drain
- Whether there is a potential health or flooding risk

This creates a more organised method of identifying waste-management problems.

2. Better prioritisation of serious problems

Not every waste report requires the same level of urgency. The proposed system therefore uses a logical priority calculation.

For example:

Flood risk = +5
Health risk = +5
Near drain = +4
High quantity = +2
People affected = +1 per person
Days waiting = +1 per day

The resulting score can then be classified as Low, Medium, High or Urgent.

This is important because it demonstrates how computational thinking can help a waste-management process make more consistent decisions.

3. Supporting better resource allocation

Once a report has been classified, the system can determine the appropriate waste-management area and assign the report to the responsible service provider.

Therefore, instead of simply recording:

"There is a lot of rubbish here."

the system produces structured information such as:

Report ID → Location → Waste Type → Priority → Responsible Area → Provider → Status.

This makes the information more useful for management and decision-making.

4. Monitoring unresolved problems

The system also includes a status-monitoring component. A report can move through stages such as:

Reported → Validated → Assigned → In Progress → Resolved.

This allows outstanding reports to be identified instead of disappearing after they are initially reported.

Open Source

The second requirement is Open Source.

For this project, Open Source means that if the proposed system were developed into actual software, its source code could be made available under an appropriate open-source licence so that other developers or organisations could study, use, modify and redistribute the software according to the licence terms.

The Open-Source Initiative explains that Open Source is based on specific licensing conditions, including access to source code, free redistribution, and permission to create modified or derived works.

How Open Source applies to this project

The proposed system is intended to solve a local problem, but the underlying logic could potentially be useful in other communities.

For example, another community could adapt the system for:

- Different waste categories
- Different priority rules
- Different service providers
- Different administrative areas
- Different reporting procedures
- Different languages
- Different local-government structures

This is one of the major advantages of an open-source approach.

Transparency

Open Source would allow developers and other stakeholders to examine how the system makes decisions.

For example, the priority algorithm could be publicly documented:

Flood Risk = +5

Health Risk = +5

Near Drain = +4

A developer or reviewer could therefore understand how the system arrives at a priority score.

Local adaptation

A system designed for Freetown may not work exactly the same way in another city.

With an open-source design, another community could modify the system's modules without having to build an entirely new system.

Similarly, the priority calculation could be modified if another municipality uses different criteria.

Collaboration

Open Source also allows different developers to contribute to improvements.

For example:

- One developer could improve the reporting module.
- Another could improve the priority algorithm.
- Another could improve the dashboard.
- Another could improve documentation.
- Another could improve data security.

Digital Public Goods (DPG)

A **Digital Public Good** can include open-source software, open data, open AI systems or open content collections that comply with relevant privacy laws and good practices, designed to avoid harm, and contribute toward the Sustainable Development Goals. The Digital Public Goods Alliance uses the DPG Standard to assess whether a solution meets these requirements.

How the Proposed System Could Meet DPG Principles

The current DPG Standard identifies nine main indicators, including SDG relevance, open licensing, clear ownership, platform independence, documentation, non-PII data extraction, privacy and applicable laws, standards and best practices, and measures addressing privacy, security and harmful content.

This project connects to these requirements as follows.

1. SDG Relevance

The system has a clear connection to SDG 11, especially Target 11.6, because its purpose is to improve the organization of waste reporting and management.

Therefore:

Project → Waste Management → Sustainable Cities → SDG 11

This gives the project a clear development objective rather than being software created without a social purpose.

Sierra Leone Community Impact

The proposed Logical Waste Collection and Community Reporting System for Freetown is designed to address a practical community challenge in Sierra Leone: the difficulty of organizing information about waste problems so that they can be reported, recorded, prioritized, assigned and monitored systematically.

The community impact should therefore focus not only on "rubbish" itself, but on how poorly coordinated waste information and reporting can affect residents, service providers, communities and the wider urban environment.

The Waste-Management Challenge in Freetown

Freetown is a densely populated urban area where effective waste management is important for maintaining healthy and sustainable communities.

A major issue that the proposed project focuses on is the coordination of waste-related information.

When a waste problem occurs, useful information may include:

- Where the problem is located
- What type of waste is involved
- How much waste is present
- How long the waste has remained
- How many people are affected
- Whether the waste is close to a drain or waterway
- Whether there is a potential health risk
- Whether there is a potential flooding risk
- Whether the problem has already been assigned for collection

Impact on Residents and Households

Residents are one of the main groups affected by waste-management problems.

When waste accumulates in a community, residents may need a reliable way to communicate the problem to the appropriate waste-management service.

The proposed system provides a structured reporting process.

Instead of an informal report such as:

"There is too much rubbish in our area."

the system can capture information such as:

Location: Community A
Waste Type: Household waste
Days Waiting: 5
Near Drain: Yes
Flood Risk: Yes
Priority: Urgent
Status: Pending

This makes the report more useful for subsequent decision-making.

Community benefit

Residents can potentially benefit from:

- More structured reporting
- Better identification of urgent problems
- Greater visibility of outstanding reports
- More consistent communication of waste problems
- Improved monitoring of reported issues

The system therefore gives residents a more organised way of contributing information about problems affecting their communities.

Impact on Businesses and Markets

Businesses and markets can also be affected by waste accumulation.

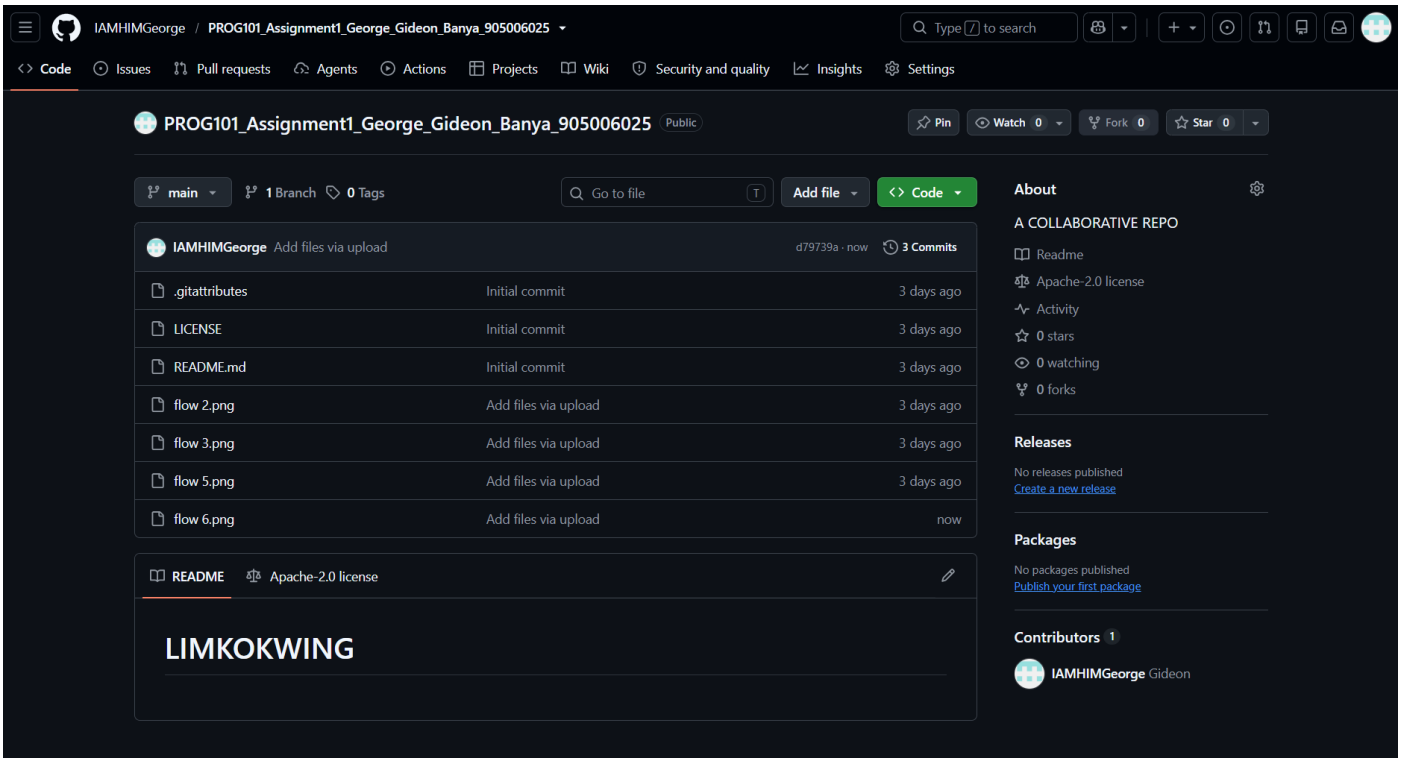
For example, waste around a market area can affect the surrounding environment and create additional pressure on waste-management services.

The proposed system can treat businesses, markets and other institutions as reporting sources, allowing their reports to be captured using the same logical structure.

For example:

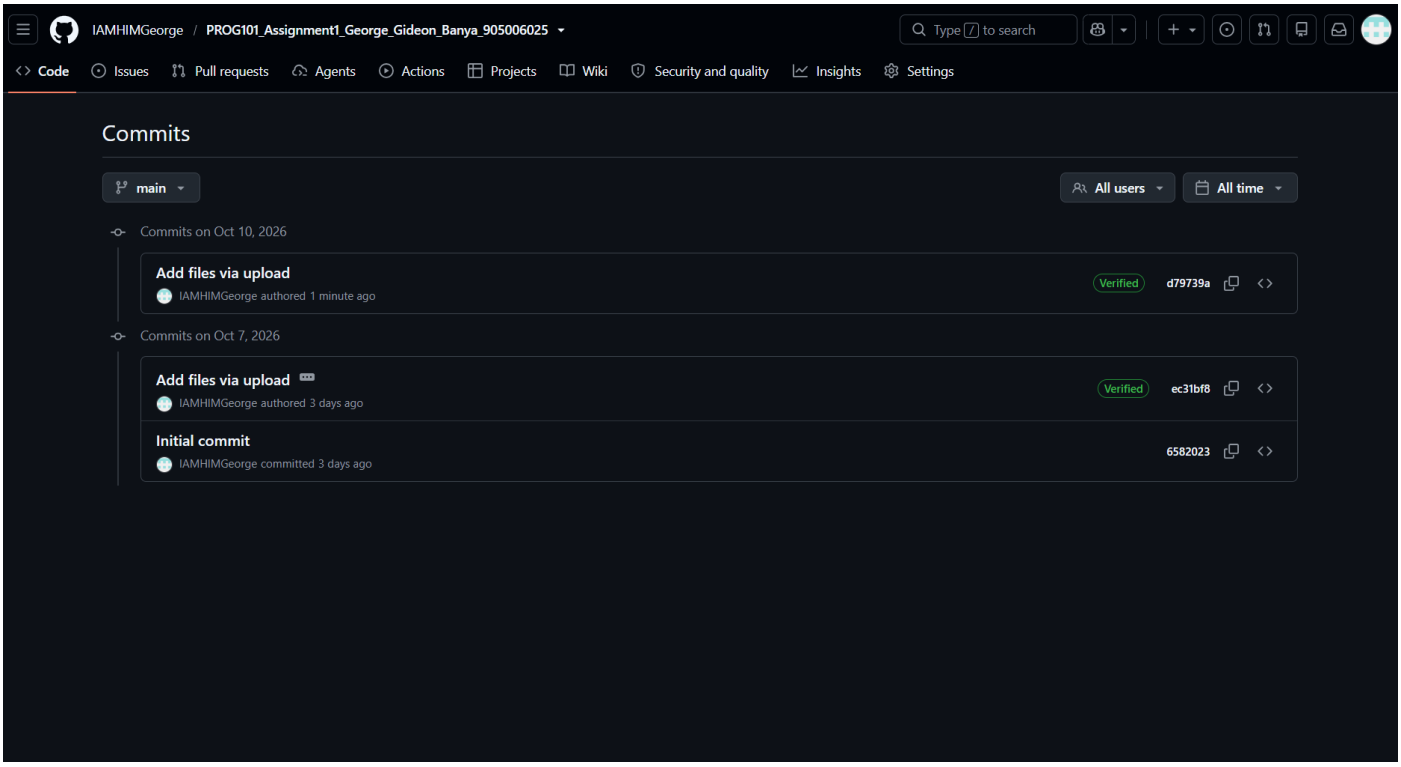
Market → Waste report → Validation → Priority assessment → Responsible area → Assignment → Status monitoring.

GITHUB REPORT OF THE PROPOSED PROJECT



A view of the repository with the diagrams of the flowcharts (flow 2, 4, 5) with the "readme.md" file and report attached

A screenshot showing the GIT commit history



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